Ziang Niu

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Education

University of Pennsylvania (Philadelphia, PA), Ph.D. in Statistics, 2028 (expected).
Research Advisor: Eugene Katsevich and Bhaswar B. Bhattacharya.

University of Pennsylvania (Philadelphia, PA), M.A. in Applied Mathematics, 2023.
Research Advisor: Eugene Katsevich and Bhaswar B. Bhattacharya.

Renmin University of China (Beijing, China), B.A. in Economic Statistics, 2021.
Thesis Advisor: Wei Li.

Fellowship and Awards

- Lawrence David Brown Best Student Paper Award (2024).
  Department of Statistics and Data Science at Wharton, UPenn.
- SIAM Annual Meeting Student Travel Award (2021).
  Society for Industrial and Applied Mathematics.

Professional Service Activities

- *Reviewer*, AoAP (1), Bernoulli (1), JASA (2), JRSS-A (1), NeurIPS (1), Statistical Science (1).
- *Organizer*, ICSA-Canada Chapter Symposium (2024).
  I organized and chaired the session "Topic in Combinatorial Inference", which included inviting speakers, coordinating the conference schedule with the speakers and hosting the session.

Presentations

Invited Seminar Presentations

- *Robust and efficient methods for single-cell CRISPR screens: PerturbPlan and spaCRT* [Slides]
  Statistics seminar at Mathematics and Statistics in McGill University, Apr. 17, 2026.
- *The permuted score test for robust differential expression analysis.* [Slides]
  Department seminar at School of Statistics and Data Science in SUFE, Dec. 31, 2025
- *Assumption-lean weak limits and tests for two-stage adaptive experiments.* [Slides]
  International Seminar on Selective Inference, Jun. 3, 2025.
- *Computationally efficient and statistically accurate conditional independence testing with spaCRT.* [Slides]
  International Seminar on Selective Inference, Nov. 4, 2024.
  Special Statistics and Biostatistics seminar at OSU, Oct. 8, 2025, in Columbus, USA.

Invited Conference Oral Presentations

- *Assumption-lean weak limits and tests for two-stage adaptive experiments.*
  International Indian Statistical Association Conference, Jun. 12, 2025.

Contributed Conference Oral Presentations

- *Detect model miscalibration via your nearest neighbor.* [Slides]
  Bernoulli-ims 11th World Congress in Probability and Statistics, Aug. 12-16, 2024, in Bochum, Germany.
- *A reconciliation between finite-sample and asymptopia-based methods in conditional independence testing.* [Slides]
  Lawrence David Brown student workshop, Mar. 22, 2024, in Philadelphia, USA.
  Joint Statistical Meeting, Aug. 5-10, 2023, in Toronto, Canada.
- *Inference for ATE using heterogeneity: generalized 2SLS and double machine learning perspectives.*
  Statistical Society of Canada Annual Meeting, May 28-31, 2023, in Ottawa, Canada.
- *Discrepancy-based Inference for Intractable Generative Models using Quasi-Monte Carlo.* [Slides]
  Lifting Inference with Kernel Embeddings, Jan. 10–14, 2022, online. [Video]
- *Estimation and inference for high-dimensional nonparametric additive instrumental-variables regression.* [Slides]
  Chinese R Conference, Nov. 20–21, 2021, in Beijing, China.
  ICSA-Canada Chapter Symposium, Jul. 8–10, 2022, in Banff, Canada.

Conference Poster Presentations

- *Discrepancy-based Inference for Intractable Generative Models using Quasi-Monte Carlo.*
  SIAM Annual Meeting, Jul. 19–23, 2021, online. [Poster]
  Paris AI Summer School, Jul. 5–9, 2021, online.

Mentorship

- Vikram Balasubramanian
  Directed Reading Program, UPenn, Sep.–Dec., 2022.
- Alexandru Lopotenco
  Undergraduate Research in Probability and Statistics, UPenn, Jan.–May., 2022.
- Ryan Jeong
  Undergraduate Research in Probability and Statistics, UPenn, Jan.–May., 2022.

Publications and Preprints

† stands for senior author; * stands for equal contribution.

[1] **Z. Niu***, J. Meier*, and F-X. Briol. Discrepancy-based Inference for Intractable Generative Models using Quasi-Monte Carlo. **Electronic Journal of Statistics**, 2022. Available on arXiv.

[2] **Z. Niu**, Y. Gu, W. Li. Estimation and inference for high-dimensional nonparametric additive instrumental-variables regression. To appear at **Electronic Journal of Statistics**, 2022+. Available on arXiv.

[3] S. Mukherjee, **Z. Niu**, S. Halder, B. B. Bhattacharya, G. Michailidis. High Dimensional Logistic Regression Under Network Dependence. To appear at **Journal of Machine Learning Research**, 2022+. Available on arXiv.

[4] **Z. Niu***, A. Chakraborty*, O. Dukes, and E. Katsevich. Reconciling model-X and doubly robust approaches to conditional independence testing. To appear at **Annals of Statistics**. Available on arXiv. [*Lawrence David Brown Best Student Paper Award at Wharton, 2024*]

[5] **Z. Niu**, B. B. Bhattacharya. Distribution-free joint independence testing and robust independent component analysis using optimal transport. Major revision at **Journal of the American Statistical Association**, 2022+. Available on arXiv.

[6] **Z. Niu**, J. Ray Choudhury, E. Katsevich. The conditional saddlepoint approximation for fast and accurate large-scale hypothesis testing. R&R at **Biometrika**, 2025+. Available on arXiv.

[7] A. Chatterjee*, **Z. Niu***, B. B. Bhattacharya. A kernel-based conditional two-sample test using nearest neighbors (With applications to calibration, regression curves and simulation-based inference). In submission, 2024. Available on arXiv.

[8] T. Barry, **Z. Niu**, E. Katsevich, X. Lin. The permuted score test for robust differential expression analysis. In submission, 2025. Available on arXiv.

[9] **Z. Niu**, Z. Ren. Assumption-lean weak limits and tests for two-stage adaptive experiments. R&R at **JRSS-B**. Available on arXiv. [*Finalist for IISA Student Paper Competition, 2025*]

[10] Z. Huang, **Z. Niu**† Semiparametric KSD test: unifying score and distance-based approaches for goodness-of-fit testing. In submission, 2025. Available on arXiv.